

Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: **C. A. Gunter and D. S. Scott**, “**Semantic Domains**,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([../papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

Work counts

- **Reuse from Mathlib — no work (12)**: poset, directed, cpo, $\perp$, monotone, continuous, Fixed-Point Theorem (1) & operator, Schröder–Bernstein (2), algebraic lattice, product, λ -notation.
- **Generalize / adapt (4)**: compact element (`IsCompactElement`, $\text{lattice} \rightarrow \text{dcpo}$), sum, lift (`WithBot/Sum`), ideal completion (`Order.Ideal`).
- **Definitions to define — new (≈ 13)**: way-below $\ll$, algebraic cpo, **domain**, bounded-complete, embedding–projection pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and D^∞ .
- **Theorems to prove (28)**: 28 of the paper’s 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

Bottom line: ≈ 13 definitions to define + 28 results to prove, on top of 12 reused Mathlib foundations.

Lean column legend: ✓ reuse Mathlib (name given) · ~ partial (Mathlib has a related or lattice-only version — generalize)
· ✗ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render $\rightarrow$ as $!$, drop `fi` ligatures, mangle math), so read them as a guide to *what* each result is. Notation: $\sqsubseteq$ order, $\sqcup$ directed sup, $\perp$ bottom, $\ll$ way-below, $K(D)$ compacts, $Fp(D)$ finitary projections.

§2 Recursive definitions of functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Def	—	poset (partially ordered set)	✓ <code>PartialOrder</code>
2.1	Def	—	directed subset M : every finite $u \subseteq M$ has an upper bound in M	✓ <code>DirectedOn / IsDirected</code>
2.1	Def	—	cpo : poset in which every directed M has a lub $\sqcup M$	✓ <code>CompletePartialOrder</code> (chains: <code>OmegaCompletePartialOrder</code>)
2.1	Def	—	bottom $\perp$ (least element)	✓ <code>OrderBot</code>
2.1	Def	—	monotone function	✓ <code>Monotone / OrderHom</code>
2.1	Def	—	continuous : monotone, $f(\sqcup M) = \sqcup f(M)$ for directed M	✓ <code>ScottContinuous</code>
2.1	Thm	1	Fixed-Point Theorem : $f : D \rightarrow D$ continuous $\implies$ least fixed point $\sqcup_n f^n(\perp)$	✓ <code>OrderHom.lfp</code> (Kleene: <code>OmegaCompletePartialOrder</code>)
2.2	Thm	2	Schröder–Bernstein for sets	✓ <code>Function.Embedding.schroederBernstein</code>
2.3	Def	—	fixed-point operator (uniform)	✓ <code>OrderHom.lfp / LawfulFix</code>
2.3	Thm	3	The standard operator is the unique uniform fixed-point operator	✗ prove

§3 Effectively presented domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	compact element x : $x \sqsubseteq \sqcup M$ (dir.) $\implies x \sqsubseteq y$ some $y \in M$; $K(D)$	~ <code>IsCompactElement</code> (lattice only $\rightarrow$ <code>dcpo</code>)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	way-below $\ll$ (approximation relation behind compactness)	✗ define
3.1	Def	—	algebraic cpo: $x = \bigsqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)	✗ define
3.1	Def	—	domain = algebraic cpo (central object)	✗ define (ScottDomain)
3.1	Def	—	bounded complete : $\perp +$ every bounded subset has a sup	✗ define
3.1	Def	—	(countably based) algebraic lattice	✓ IsCompactlyGenerated (+CompleteLattice)
3.1	Def	—	embedding-projection pair (g, f)	✗ define
3.1	Def	—	projection ; finitary projection p : $p \circ p = p \sqsubseteq \text{id}$, $\text{im}(p)$ a domain	✗ define
3.1	Def	—	normal subposet / substructure	✗ define
3.1	Lem	4	$\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least	✗ prove
3.1	Lem	5	p finitary projection $\implies$ compacts of $\text{im}(p)$ are $\{p(x) : x \in K(D)\}$	✗ prove
3.1	Thm	6	Isomorphism: normal substructures $\cong \text{Fp}(D)$ (finitary projections)	✗ prove
3.1	Thm	7	D, E bounded-complete domains $\implies D \rightarrow E$ bounded-complete domain	✗ prove
3.2	Def	—	effective presentation $d : \mathbb{N} \rightarrow K(D)$; effectively presented domain	✗ define

§4 Operators and functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.1	Def	—	product $D \times E$	✓ Prod (order/cpo instances)
4.2	Def	—	Church's λ-notation (continuous abstraction)	✓ OrderHom / ω CPO ContinuousHom
4.3	Def	—	smash product $D \otimes E$	✗ define
4.4	Def	—	sum $D + E$; lift $D \perp$	~ Sum / WithBot, Part (partial)
4.x	Lem	8	Currying/iso laws for function spaces over D, E, F	✗ prove
4.x	Lem	9	Product/function-space iso laws over D, E, F	✗ prove
4.5	Lem	10	D, E bounded complete $\implies \rightarrow, \times, \otimes, +, () \perp$ bounded complete	✗ prove
4.5	Thm	11	Ideal completion of a countable pre-order is a domain (all domains so arise)	~ Order.Ideal exists $\rightarrow$ prove
4.5	Thm	12	Initiality of a continuous algebra satisfying axioms T	✗ prove
4.5	Lem	13	D bounded complete $\implies$ powerdomains $D], D[$ bounded complete	✗ prove
4.5	Thm	14	Equivalent characterizations of an (algebraic/BC) domain	✗ prove

§5 Powerdomains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
5.1	Def	—	powerdomain (non-deterministic outcomes)	✗ define
5.2	Def	—	Hoare (lower), Smyth (upper), Plotkin (convex) powerdomains	✗ define
5.3	—	—	Universal & closure properties (see Lem 13, 28, 30)	✗ prove

§6 Bifinite (SFP) domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.1	Def	—	Plotkin order / bifinite (SFP) domain: bilimit of finite pointed posets	✗ define
6.1	Prop	15	Every bounded-complete domain is bifinite	✗ prove
6.1	Thm	16	D bifinite $\implies \text{Fp}(D)$ is an algebraic lattice	✗ prove
6.2	Lem	17	D, E bifinite $\implies \rightarrow, \times, \otimes, +, ()^\perp$ bifinite (incl. function space)	✗ prove
6.2	Thm	18	If D and $D \rightarrow D$ are domains, then D is bifinite	✗ prove
6.2	Lem	19	closure $r: D \rightarrow D$ ($r \circ r = r \sqcup \text{id}$) $\implies \text{im}(r)$ is a domain	✗ prove
6.2	Lem	20	D domain $\implies \text{Fc}(D)$ (finitary closures) is a cpo	✗ prove

§7 Recursive definitions of domains (universal domain, D_∞)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Def	—	recursive domain equation; universal domain / D_∞ (inverse limit)	✗ define
7	Thm	21	F representable over cpo $U \implies$ a domain D with $D \cong F(D)$	✗ prove
7	Thm	22	Any countably-based algebraic lattice L : a closure $r : P(\mathbb{N}) \rightarrow L$	$\sim \text{IsCompactlyGenerated} \rightarrow$ prove
7	Lem	23	The function-space operator is representable over $P(\mathbb{N})$	✗ prove
7	Lem	24	U cpo; $\times$ and $\rightarrow$ representable over $U \implies$ (setup for universality)	✗ prove
7	Thm	25	U non-trivial domain representing $\times, \rightarrow \implies U$ universal	✗ prove
7	Thm	26	Any signature $(s_1, \dots, s_n)$: combinators $F_1, \dots, F_n$ solving the equations	✗ prove
7	Thm	27	Any bounded-complete D : a projection of the universal domain onto D	✗ prove
7	Lem	28	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ representable over U	✗ prove
7	Thm	29	D bifinite $\implies D^+$ bifinite; solving $D \cong D^+$	✗ prove
7	Lem	30	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ p-representable over universal bifinite V	✗ prove

Tally (matched): ✓ reuse Mathlib $\approx$ **12** (poset, directed, cpo, $\perp$, monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product, λ -notation) · $\sim$ partial $\approx$ **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) · $\times$ define / prove $\approx$ **44** — the entire domain / $\ll$ / bounded-complete / bifinite / powerdomain / D^∞ development.

First target unchanged: WayBelow. lean ($\ll$, $\times$), since compact, algebraic, domain, and bounded complete all sit on it.